Case 02 · The final comparison and the decision

The stage after architecture. Until now each idea was tested on its own, against its own requirements. Here the two that reached architecture face each other and are asked the same question: the question that sent the whole Case on its way.

The deciding question

The central problem we chose in stage 2 is "a crisis in slow motion": the damage of high interest rates seeps in slowly, and then arrives all at once. History (British pension funds in 2022, SVB in 2023) taught that the problem was never a lack of information, but that no one connected it in time.

So the deciding question is: which of them lets us see the fast part of the crisis, before it arrives?

The fast part always came from what could not be seen: hidden leverage, hidden exposure, and a loop of forced selling no one knew was primed. Both survivors touch exactly that, each from a different direction.

Three rounds

Round one: the mechanism. Does the core work?

- **Fire-Sale Cascade Simulator:** yes. The POC reproduced the British loop (a 0.6-point shock amplified 1.9 times), showed that an intervention breaks it, and that a world without leverage stays stable. A thousand runs in under a second.
- **Private Exposure Registry:** yes. The POC combined the data of 500 participants, exactly, without any server seeing a single data point, in 0.11 seconds. And the spec built the core on a standard in development at the IETF and code already running in production.

A draw. And no surprise: both passed POC, spec and architecture. A working core is an entry requirement, not an advantage.

Round two: the input. Where does the data the answer depends on come from?

- **Fire-Sale Cascade Simulator:** from outside. The sweep in feature 8 measures it itself: between average leverage of 4 and 5.5 the outcome moves from "calm" to "near-certain spiral". The system does not and cannot collect that leverage. It can guess a range, or wait for someone to give it one. In the component diagram it is a red dashed box with a question mark, and no component fills it.
- **Private Exposure Registry:** from inside. Leverage and exposures come in from the participants themselves, through the system, and leave it as a verified picture. Its three external dependencies (GLEIF, drand, existing regulatory reporting) serve identity and audit, not the answer.

Here the paths diverge. One of them knows how to compute what would happen if we knew. The other knows.

Round three: crisis day. What is on the regulator's desk?

- **Fire-Sale Cascade Simulator:** a fan of scenarios, each beginning with the word "if". "If average leverage is 4, the market will hold. If it is 5, probably not." Very useful for thinking, but it does not answer the question "what is the situation now".
- **Private Exposure Registry:** a picture of now, for example (in the POC's synthetic numbers): "17% of exposure carries rates above 5%, and a third of funds are leveraged above 4 times". Verified against commitments, with no entity exposed, and with proof in the log that it was not altered.

A scenario can be argued with. A verified number is hard to explain away.

The decision

The winner: Private Exposure Registry.

It is the one that closes the loop end to end: the decisive input enters through it, passes through it, and leaves it verified, all within its own boundaries. It does not ask anyone to supply the missing data. It collects them.

Fire-Sale Cascade Simulator falls at architecture. Not because of a flaw: everything inside its boundaries works well. It falls because the closure check showed that its central promise, to say something about today's market, passes through a box it does not have.

The twist

Remember the red box in the Fire-Sale diagram, "exposure data for today's market"? That is exactly what the Registry built. It is not the simulator's competitor; it is its missing piece.

And the link runs the other way too: the Registry's spec ends with the sentence "it shows the picture, not what will happen to it". That is exactly what the simulator knows how to do. And feature 10 of the Registry includes an export of the leverage and exposure distributions "in a format an external model can load as a distribution instead of inventing one".

So why does the Registry win, and not the other way round? Because the dependency runs in one direction. The Registry is fully useful without the simulator: a regulator who sees that a third of the market is leveraged above 4 times already knows there is a problem. The simulator without the Registry answers "it depends". Whoever holds the input holds the answer.

What the winner does not solve (honestly)

- **A participant who lies:** the proofs guarantee that a number is in range, not that it is true. Random audits deter, but the system is no more reliable than the regulatory reporting beneath it.
- **Three authorities:** the system requires three supervisory authorities to run a server and agree on a catalog. That is an organisational and political challenge, not a technical one, and it has no solution in code.
- **What is unsupervised:** the system sees only entities that have an LEI and a regulator. What happens outside the supervised system stays in the dark.

What we learned from this Case

In Case 01 the lesson was "a system that sees without the ability to act is a report". In Case 02 the lesson is close, but different: a system that computes without knowing is a scenario. An excellent simulation fed with a guess produces an excellent guess. The real value lay in the data no one saw, and in a way to see them without exposing anyone.

And the question that stays open: if data can be combined without being seen, why do authorities still not do it, and who should go first?